BASIC PYTHON

------VARIABLES------

In [1]:
```
import sys
sys.version
```

Out[1]:
```
'3.13.9 | packaged by Anaconda, Inc. | (main, Oct 21 2025, 19:09:58) [MSC v.1929 6
4 bit (AMD64)]'
```

In [2]:
```
v = 5
v
```

Out[2]:
```
5
```

In [3]:
```
5 = v
```

```
  Cell In[3], line 1
    5 = v
    ^
SyntaxError: cannot assign to literal here. Maybe you meant '==' instead of '='?
```

In [ ]:
```
5v = 6
5v
```

In [ ]:
```
v5 = 6
v5
```

In [4]:
```
Alfiya = 20
ALFIYA
```

```
---------------------------------------
NameErrorTraceback (most recent call last)
Cell In[4], line 2
      1 Alfiya = 20
----> 2 ALFIYA

NameError: name 'ALFIYA' is not defined
```

In [ ]:
```
Alfiya
```

In [ ]:
```
Alfiya = 34
```

In [5]:
```
Alfiya* = 34
```

```
  Cell In[5], line 1
    Alfiya* = 34
            ^
SyntaxError: invalid syntax
```

In [6]:
```
x_train = 80
x_train
```

Out[6]:
```
80
```

In [7]:
```
import keyword
keyword.kwlist
```

Out[7]:
```
['False',
 'None',
 'True',
 'and',
 'as',
 'assert',
 'async',
 'await',
 'break',
 'class',
 'continue',
 'def',
 'del',
 'elif',
 'else',
 'except',
 'finally',
 'for',
 'from',
 'global',
 'if',
 'import',
 'in',
 'is',
 'lambda',
 'nonlocal',
 'not',
 'or',
 'pass',
 'raise',
 'return',
 'try',
 'while',
 'with',
 'yield']
```

In [8]:
```
len(keyword.kwlist)
```

Out[8]:
```
35
```

In [9]:
```
for = 67
```

```
  Cell In[9], line 1
    for = 67
        ^
SyntaxError: invalid syntax
```

In [10]:
```
For = 67
For
```

Out[10]: 67

In [11]:
```
False = 67
```

```
  Cell In[11], line 1
    False = 67
    ^
SyntaxError: cannot assign to False
```

In [12]:
```
false = 67
false
```

Out[12]: 67

- variable is case sensistive - var doesnot start with digit but end with digit - special chare will decare as variable except _ - keywords reserved word - 35 reserve words | True, False , None - keyword alwasy assigned with Cap

PYTHON DATATYPES

variable naene = value variable == we are discusse today and we completed value == data type python data types 1- int - value without decimal 2- float - value with decimal 3- bool - True or False 4- complex - Represent no with real and imagery part 5- string - It used to store the character

(1) INTEGER DATATYPE

In [13]:
```
4   #int without variable
```

Out[13]: 4

In [14]:
```
10
```

Out[14]: 10

In [15]:
```
4 +10
```

Out[15]: 14

In [16]:
```
5 - 6
```

Out[16]: -1

In [17]:
```
i = 5 #int with varibale
i
```

Out[17]: 5

In [18]:
```
i = 56
i
```

Out[18]: 56

In [19]:
```
id(i)
```

Out[19]: 140724644326024

In [20]:
```
type(i)
```

Out[20]: int

In [21]:
```
int(True)
```

Out[21]: 1

In [22]:
```
int(False)
```

Out[22]: 0

(2) FLOAT DATATYPE

In [23]:
```
petrol_price = 110.45
petrol_price
```

Out[23]: 110.45

In [24]:
```
gold_price = 11000.56
gold_price
```

Out[24]: 11000.56

In [25]:
```
f2 = 2e0
f2
```

Out[25]: 2.0

In [26]:
```
type(f2)
```

Out[26]: float

In [27]:
```
f3 = 2.5e2
f3
```

Out[27]: 250.0

In [28]:
```
f4 = 3.4E3
f4
```

Out[28]: 3400.0

(3) BOOLEAN (BOOL) DATATYPE

In [29]:
```
True
```

Out[29]:
```
True
```

In [30]:
```
False
```

Out[30]:
```
False
```

In [31]:
```
None
```

In [32]:
```
print(True + True)
print(True + False)
print(True - True)
print(False - False)
```

```
2
1
0
0
```

In [33]:
```
10 / 2 #float division
```

Out[33]:
```
5.0
```

In [34]:
```
10 // 2
```

Out[34]:
```
5
```

In [35]:
```
False / True
```

Out[35]:
```
0.0
```

In [36]:
```
False // True
```

Out[36]:
```
0
```

In [37]:
```
True / False
```

```
---------------------------------------------
ZeroDivisionErrorTraceback (most recent call last)
Cell In[37], line 1
----> 1 True / False

ZeroDivisionError: division by zero
```

(4) COMPLEX DATATYPE

FORMATE of the complex -------> a + bj

In [38]:
```
c = 10 + 20j
c
```

Out[38]:
```
(10+20j)
```

In [39]:
```
type(c)
```

Out[39]:
```
complex
```

In [40]:
```
d = 20 + 30j
d
```

Out[40]:
```
(20+30j)
```

In [41]:
```
print(c)
print(d)
```

```
(10+20j)
(20+30j)
```

In [42]:
```
print(type(i))
```

```
<class 'int'>
```

In [44]:
```
type(c)
```

Out[44]:
```
complex
```

In [45]:
```
c
```

Out[45]:
```
(10+20j)
```

In [46]:
```
c.real
```

Out[46]:
```
10.0
```

In [47]:
```
c.imag
```

Out[47]:
```
20.0
```

(5) STRING DATATYPE

In [48]:
```
welcome
```

```
---------------------------------------------
NameErrorTraceback (most recent call last)
Cell In[48], line 1
----> 1 welcome

NameError: name 'welcome' is not defined
```

In [49]:
```
welcome = 7
welcome
```

Out[49]: 7

In [50]:
```
khan = 5
khan
```

Out[50]: 5

python string must assinged '' | "" | ''' '''

In [51]:
```
welcome to the jupyter Notebook
```

```
  Cell In[51], line 1
    welcome to the jupyter Notebook
            ^
SyntaxError: invalid syntax
```

In [52]:
```
"welcome to the jupyter Notebook"
```

Out[52]: 'welcome to the jupyter Notebook'

In [53]:
```
'''welcome
to
the
jupyter
Notebook'''
```

Out[53]: 'welcome \nto \nthe\njupyter\nNotebook'

In [54]:
```
str1 = "HELLO PYTHON"
print(str1)
```

```
HELLO PYTHON
```

In [55]:
```
mystr = 'Hello World' # Define string using single quotes
print(mystr)
```

```
Hello World
```

In [56]:
```
mystr = "Hello World" # Define string using double quotes
print(mystr)
```

```
Hello World
```

In [57]:
```
mystr = '''Hello
World ''' # Define string using triple quotes
print(mystr)
```

```
Hello
World
```

In [58]:
```
mystr = """Hello
World""" # Define string using triple quotes
print(mystr)
```

```
Hello
World
```

In [59]:
```
mystr = ('Happy '
'Monday ' 'Everyone')
print(mystr)
```

```
Happy Monday Everyone
```

In [60]:
```
mystr2 = 'Woohoo '
mystr2 = mystr2*5
mystr2
```

Out[60]:
```
'Woohoo Woohoo Woohoo Woohoo Woohoo '
```

In [61]:
```
len(mystr2) # Length of string
```

Out[61]:
```
35
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: